Laporan P.ASD

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2.2.1 Praktikum Pemilihan

```
package PraktikumASD.jobsheet1;
import java.util.Scanner;

public class praktikumPenilaian {
    static double[] nilai = new double[4];
    static double[] outerlayer(double nilai[]){
        Scanner sc = new Scanner(System.in);

        while (true) {
            System.out.print("Masukan Nilai Tugas: ");
            int counter = 0;

            nilai[0] = sc.nextDouble();
            if (nilai[0] <= 100 && nilai[0] >= 0) {
                counter++;
            }

            System.out.print("Masukan Nilai Kuis: ");
            nilai[1] = sc.nextDouble();
            if (nilai[1] <= 100 && nilai[1] >= 0) {
                counter++;
            }

            System.out.print("Masuka Nilai UTS: ");
            nilai[2] = sc.nextDouble();
            if (nilai[2] <= 100 && nilai[2] >= 0) {
                counter++;
            }

            System.out.print("Masukan Nilai UAS: ");
            nilai[3] = sc.nextDouble();
            if (nilai[3] <= 100 && nilai[3] >= 0) {
                counter++;
            }

            if (counter == 4) {
                break;
            } else {
                System.out.println("================================");
                System.out.println("================================");
                System.out.println("nilai tidak valid");
                System.out.println("================================");
                System.out.println("================================");
            }
        }
            return nilai;
        }


    static void pengujian(double[] nilai) {
```

```
        double nilaiAkhir = 0;
        String nilaiHuruf;
        double tugas = nilai[0] * 0.2;
        double kuis = nilai[1] * 0.2;
        double uts = nilai[2] * 0.3;
        double uas = nilai[3] * 0.3;

        double total = tugas + kuis + uts + uas;
        if (total > 80 && total <= 100) {
            nilaiAkhir = total;
            nilaiHuruf = "A";
        } else if (total > 73 && total <= 80) {
            nilaiAkhir = total;
            nilaiHuruf = "B+";
        } else if (total > 65 && total <= 73) {
            nilaiAkhir = total;
            nilaiHuruf = "B";
        } else if (total > 60 && total <= 65) {
            nilaiAkhir = total;
            nilaiHuruf = "C+";
        } else if (total > 50 && total <= 60) {
            nilaiAkhir = total;
            nilaiHuruf = "C";
        } else if (total > 39 && total <= 50) {
            nilaiAkhir = total;
            nilaiHuruf = "D";
        } else if (total <= 39) {
            nilaiAkhir = total;
            nilaiHuruf = "E";
        } else {
            nilaiAkhir = total;
            nilaiHuruf = "Bugg";
        }

        System.out.println("nilai akhir: " + nilaiAkhir);
        System.out.println("nilai huruf: " + nilaiHuruf);
    }
    public static void main(String[] args) {
        System.out.println("Program Menghitung Tugas Akhir");
        System.out.println("================================");
        outerlayer(nilai);
        pengujian(nilai);
        System.out.println("================================");
        System.out.println("================================");
    }
}
```

Hasil running :

```
PS E:\GitHub Utilities>  e:; cd 'e:\GitHub Utilities'; & 'C:\Program
essages' '-cp' 'C:\Users\ironPC\AppData\Roaming\Code\User\workspaceS
bin' 'PraktikumASD.jobsheet1.praktikumPenilaian'
Program Menghitung Tugas Akhir
================================
Masukan Nilai Tugas: 80
Masukan Nilai Kuis: 90
Masuka Nilai UTS: 75
Masukan Nilai UAS: 87
nilai akhir: 82.6
nilai huruf: A
================================
================================
PS E:\GitHub Utilities>
```

2.3.1 Praktikum Perulangan

```java
package PraktikumASD.jobsheet1;

import java.util.Scanner;

public class praktikumPerulangan {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double nim = 0.0, n;
        System.out.print("Masukan Nim Anda: ");
        nim = sc.nextDouble();
        n = nim % 100;

        System.out.println(n);

        for (int i = 1; i <= n; i++) {
            if (i == 10 || i == 15) {

            } else if (i % 3 == 0){
                System.out.print("# ");
            } else if (i % 2 != 1){
                System.out.print("* ");
            } else {
                System.out.print(i + " ");
            }
        }
    }
}
```

Hasil Running :

```
PS E:\GitHub Utilities> e:; cd "e:\GitHub Utilities"; & "C:\Program Files\Java\jdk-25\bin\java.exe" "--enable-preview" "-XX:+ShowCodeDetailsInExceptionM
essages" "-cp" "C:\Users\ironPC\AppData\Roaming\Code\User\workspaceStorage\4d0d4e5c7717b25a600b1d93b14d0758\redhat.java\jdt_ws\GitHub Utilities_6050a177
bin" "PraktikumASD.jobsheet1.praktikumPerulangan"
Masukan Nim Anda: 254107020079
79.0
* 2 # 4 * # * 8 # * # * 14 16 * # * 20 # 22 * # * 26 # 28 * # * 32 # 34 * # * 38 # 40 * # * 44 # 46 * # * 50 # 52 * # * 56 # 58 * # * 62 # 64 * # * 68 #
70 * # * 74 # 76 * # *
PS E:\GitHub Utilities>
```

2.4.1 Praktikum Array

```
package PraktikumASD.jobsheet1;

import java.util.Scanner;

public class praktikumArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("==========================================
=");
        System.out.println("Program Menghitung IP Semester");
        System.out.println("==========================================
=");

        System.out.print("Berapa mata kuliah yang kamu pelajari: ");
        int mataKuliah = sc.nextInt();
        sc.nextLine();

        String arrayMK[] = new String [mataKuliah];
        double arrayNilai[] = new double [mataKuliah];
        String arrayHuruf[] = new String[mataKuliah];
        double arraySKS[] = new double[mataKuliah];
        double arrayBobot[] = new double[mataKuliah];
         for (int i = 0; i < mataKuliah; i++) {
            System.out.print("Apa MK ke-" + (i+1) + " mu: ");
            arrayMK[i] = sc.nextLine();

            System.out.print("Berapa SKS untuk pelajaran " + arrayMK[i] + "mu: ");
            arraySKS[i] = sc.nextDouble();

            System.out.print("Masukan Nilai Angka untuk MK " + arrayMK[i] + ": ");
            arrayNilai[i] = sc.nextDouble();
            sc.nextLine();

        }

        for (int i = 0; i < arrayNilai.length; i++) {
        if (arrayNilai[i] > 80 && arrayNilai[i] <= 100) {
            arrayBobot[i] = 4;
            arrayHuruf[i] = "A";
        } else if (arrayNilai[i] > 73 && arrayNilai[i] <= 80) {
            arrayBobot[i] = 3.5;
            arrayHuruf[i]= "B+";
        } else if (arrayNilai[i] > 65 && arrayNilai[i] <= 73) {
```

```
            arrayBobot[i] = 3;
            arrayHuruf[i] = "B";
        } else if (arrayNilai[i] > 60 && arrayNilai[i] <= 65) {
            arrayBobot[i] = 2.5;
            arrayHuruf[i] = "C+";
        } else if (arrayNilai[i] > 50 && arrayNilai[i] <= 60) {
            arrayBobot[i] = 2;
            arrayHuruf[i] = "C";
        } else if (arrayNilai[i] > 39 && arrayNilai[i] <= 50) {
            arrayBobot[i] = 1.5;
            arrayHuruf[i] = "D";
        } else {
            arrayBobot[i] = 1;
            arrayHuruf[i] = "E";
        }

    }
    float sigma = 0;
    float alpha = 0;
    for (int i = 0; i < mataKuliah; i++) {
        sigma += (arrayBobot[i] * arraySKS[i]);
    }

    for (int i = 0; i < arraySKS.length; i++) {
        alpha += arraySKS[i];
    }
    float IP = sigma / alpha;


        System.out.println("==========================================
===");
        System.out.println("Hasil Konversi Nilai ");
        System.out.println("==========================================
===");

        System.out.println("MK\t\t\t\tNilai Angka\tNilai Huruf\tBobot");
        for (int i = 0; i < mataKuliah; i++) {
            System.out.printf("%-30s %.2f\t\t%s\t\t%.2f%n", arrayMK[i], arrayNilai[i],
arrayHuruf[i], arrayBobot[i]);
        }

        System.out.println("==========================================")
;
        System.out.println("IP: " + IP);
    }
}
```

Hasil Running:

```
bin' 'PraktikumASD.jobsheet1.praktikumArray'
==========================================
Program Menghitung IP Semester
==========================================
Berapa mata kuliah yang kamu pelajari: 5
Apa MK ke-1 mu: a
Berapa SKS untuk pelajaran amu: 4
Masukan Nilai Angka untuk MK a: 70,5
Apa MK ke-2 mu: b
Berapa SKS untuk pelajaran bmu: 3,5
Masukan Nilai Angka untuk MK b: 80
Apa MK ke-3 mu: c
Berapa SKS untuk pelajaran cmu: 2
Masukan Nilai Angka untuk MK c: 90
Apa MK ke-4 mu: d
Berapa SKS untuk pelajaran dmu: 3,5
Masukan Nilai Angka untuk MK d: 90
Apa MK ke-5 mu: e
Berapa SKS untuk pelajaran emu: 2
Masukan Nilai Angka untuk MK e: 70
============================================
Hasil Konversi Nilai
============================================
MK                          Nilai Angka     Nilai Huruf     Bobot
a                           70,50           B               3,00
b                           80,00           B+              3,50
c                           90,00           A               4,00
d                           90,00           A               4,00
e                           70,00           B               3,00
==========================================
IP: 3.4833333
```

2.5.1 Praktikum Fungsi

```
package PraktikumASD.jobsheet1;

public class praktikumRoyalGarden {
    static int arrayHarga[] = {75000, 50000, 60000, 10000};
    static String arrayBunga[] = {"aglonema", "keladi", "alocasia", "mawar"};
    static int arrayPendapatan[] = new int[4];
    static int arrayStok[][] = {{10, 5, 15, 7},
                                {6, 11, 9, 12},
                                {2, 10, 10, 5},
                                {5, 7, 12, 9}};
    public static void main(String[] args){

        rumusPendapatan(arrayPendapatan);

    }
    public static void rumusPendapatan(int arrayPendapatan[]){

        for (int i = 0; i < arrayStok.length; i++) {
            int counter = 0;
            for (int j = 0; j < arrayStok[i].length; j++) {
                counter += arrayStok[i][j] * arrayHarga[j];
            }
            arrayPendapatan[i] = counter;
            if (arrayPendapatan[i] > 1500000) {
```

```
                System.out.println("Toko ke- " + (i+1) + " adalah " + arrayPendapatan[i] + "
Pendapatanmu : 'Sangat Baik'");
            } else {
                System.out.println("Toko ke- " + (i+1) + " adalah " + arrayPendapatan[i] + "
Pendapatanmu : 'perlu dievaluasi'");
            }
        }
    }
}
```

Hasil Running:

```
oaming\Code\User\workspaceStorage\4d9d4e5c7717b25a600b1d93b14
Toko ke- 1 adalah 1970000 Pendapatanmu : 'Sangat Baik'
Toko ke- 2 adalah 1660000 Pendapatanmu : 'Sangat Baik'
Toko ke- 3 adalah 1300000 Pendapatanmu : 'perlu dievaluasi'
Toko ke- 4 adalah 1535000 Pendapatanmu : 'Sangat Baik'
PS E:\GitHub Utilities>
```

Tugas 1:

```
package PraktikumASD.jobsheet1;

import java.util.Scanner;

public class tugas1Kota {
    private static char[] kode = {'a', 'b', 'd', 'e', 'f', 'g', 'h', 'l', 'n', 't'};
    private static char[][] kota =      {{'B','A','N','T','E','N'},
                                        {'J','A','K','A','R','T','A'},
                                        {'B','A','N','D','U','N','G'},
                                        {'C','I','R','E','B','O','N'},
                                        {'B','O','G','O','R'},
                                        {'P','E','K','A','L','O','N','G','A','N'},
                                        {'S','E','M','A','R','A','N','G'},
                                        {'S','U','R','A','B','A','Y','A'},
                                        {'M','A','L','A','N','G'},
                                        {'T','E','G','A','L'}};
    public static void main(String[] args) {
        pencarianKode();
    }
    static void pencarianKode(){
        Scanner sc = new Scanner(System.in);
        int counter = 0;
        boolean p = true;
        while (p) {
            System.out.print("masukan kode kota: ");
            char code = sc.next().charAt(0);
            for (int i = 0; i < kode.length; i++) {
                if (code == kode[i]) {
                    counter = i;
                    System.out.print("kodemu adalah kota ");
                    for (int j = 0; j < kota[counter].length; j++) {
                        System.out.print(kota[counter][j]);
                    }
```

```
                    p = false;

                }
            }
        }


    }
}
```

Hasil running:

Tugas 2

```
package PraktikumASD.jobsheet1;
import java.util.Scanner;

public class tugas2ProgramMahasiswa {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Masukan Jadwal: ");
        int n = sc.nextInt();
        sc.nextLine();

        String[][] arrayJadwalMahasiswa = new String[n][4];

        int pos = 0;
        System.out.println("Selamat Datang Di Menu Pencatatan Jadwal Mahasiswa");

        while (true) {
            System.out.println("pilih menu: \t 1: input data\t 2: tabel jadwal\t 3: cari data sesuai
            hari\t 4: cari data sesuai matakuliah");
             System.out.print("===");
             pos = sc.nextInt();

    switch (pos) {
        case 1:
            inputJadwalKuliah(arrayJadwalMahasiswa);
            break;

            case 2:
                tabelJadwal(arrayJadwalMahasiswa);
                break;
```

```
                    case 3:
                        mencariHariKuliah(arrayJadwalMahasiswa);
                        break;

                        case 4:
                            mencariJadwalKuliah(arrayJadwalMahasiswa);
                            break;

                            default:
                                System.out.println("kode invalid.");
                                continue;
                            }
                            }
    }

    static void mencariJadwalKuliah(String[][] arrayJadwalMahasiswa) {
    Scanner sc = new Scanner(System.in);
    boolean ditemukan = false;

    System.out.print("Masukkan nama mata kuliah yang dicari: ");
    String key = sc.nextLine();

    System.out.println("\nHasil pencarian mata kuliah: " + key);
    System.out.println("==================================================
==========================");
    System.out.printf("%-25s %-14s %-20s %-15s\n",
            "Nama Mata Kuliah", "Ruang", "Hari Kuliah", "Jam Kuliah");
    System.out.println("==================================================
==========================");

    for (int i = 0; i < arrayJadwalMahasiswa.length; i++) {
        if (arrayJadwalMahasiswa[i][0].equalsIgnoreCase(key)) {
            System.out.printf("%-25s %-14s %-20s %-15s\n",
                    arrayJadwalMahasiswa[i][0],
                    arrayJadwalMahasiswa[i][1],
                    arrayJadwalMahasiswa[i][2],
                    arrayJadwalMahasiswa[i][3]);
            ditemukan = true;
        }
    }

    if (!ditemukan) {
        System.out.println("Mata kuliah tidak ditemukan.");
    }

    System.out.println("==================================================
==========================");
    }

    static void mencariHariKuliah(String[][] arrayJadwalMahasiswa){
            Scanner sc = new Scanner(System.in);
            boolean ditemukan = false;

            System.out.print("Hari: ");
            String key = sc.nextLine();

            System.out.println("\nHasil pencarian jadwal hari " + key + ":");
            System.out.println("==================================================
==============================");
```

```
        System.out.printf("%-25s %-14s %-20s %-15s\n",
                "Nama Mata Kuliah", "Ruang", "Hari Kuliah", "Jam Kuliah");
        System.out.println("==============================================
==============================");

        for (int i = 0; i < arrayJadwalMahasiswa.length; i++) {
            if (arrayJadwalMahasiswa[i][2].equalsIgnoreCase(key)) {
                System.out.printf("%-25s %-14s %-20s %-15s\n",
                        arrayJadwalMahasiswa[i][0],
                        arrayJadwalMahasiswa[i][1],
                        arrayJadwalMahasiswa[i][2],
                        arrayJadwalMahasiswa[i][3]);
                ditemukan = true;
            }
        }

        if (!ditemukan) {
            System.out.println("Tidak ada jadwal kuliah pada hari tersebut.");
        }

    System.out.println("==============================================
==============================");
    }

    static void tabelJadwal(String[][] arrayJadwalMahasiswa) {
        System.out.println();
        System.out.printf("%-25s %-14s %-20s %-15s\n",
                "Nama Mata Kuliah", "Ruang", "Hari Kuliah", "Jam Kuliah");

        System.out.println("==============================================
==============================");

        for (int i = 0; i < arrayJadwalMahasiswa.length; i++) {
            System.out.printf("%-25s %-14s %-20s %-15s\n",
                    arrayJadwalMahasiswa[i][0],
                    arrayJadwalMahasiswa[i][1],
                    arrayJadwalMahasiswa[i][2],
                    arrayJadwalMahasiswa[i][3]);
        }

        System.out.println("==============================================
==============================");
    }

    static void inputJadwalKuliah(String[][] arrayJadwalMahasiswa) {
        Scanner sc = new Scanner(System.in);

        for (int i = 0; i < arrayJadwalMahasiswa.length; i++) {
            System.out.println("Input Data jadwal ke-" + (i + 1));

            System.out.print("Masukan Nama mata kuliah: ");
            arrayJadwalMahasiswa[i][0] = sc.nextLine();

            System.out.print("Ruang Kuliah: ");
            arrayJadwalMahasiswa[i][1] = sc.nextLine();

            System.out.print("Masukan Hari kuliah: ");
            arrayJadwalMahasiswa[i][2] = sc.nextLine();
```

```
            System.out.print("Jam kuliah: ");
            arrayJadwalMahasiswa[i][3] = sc.nextLine();

            System.out.println();
        }
    }
}
```

Hasil Running:

```
Masukan Jadwal: 3
Selamat Datang Di Menu Pencatatan Jadwal Mahasiswa
pilih menu:     1: input data   2: tabel jadwal        3: cari data sesuai hari       4: cari data sesuai matakuliah
===1
Input Data jadwal ke-1
Masukan Nama mata kuliah: daspro
Ruang Kuliah: rt1\
Masukan Hari kuliah: senin
Jam kuliah: 8 - 12

Input Data jadwal ke-2
Masukan Nama mata kuliah: agama
Ruang Kuliah: rt 4
Masukan Hari kuliah: jumat
Jam kuliah: 7 - 9

Input Data jadwal ke-3
Masukan Nama mata kuliah: pcx
Ruang Kuliah: lpr 2
Masukan Hari kuliah: rabu
Jam kuliah: 10 - 14

pilih menu:     1: input data   2: tabel jadwal        3: cari data sesuai hari       4: cari data sesuai matakuliah
===2

Nama Mata Kuliah         Ruang          Hari Kuliah         Jam Kuliah
==========================================================================
daspro                   rt1\           senin               8 - 12
agama                    rt 4           jumat               7 - 9
pcx                      lpr 2          rabu                10 - 14
==========================================================================
pilih menu:     1: input data   2: tabel jadwal        3: cari data sesuai hari       4: cari data sesuai matakuliah
===3
Hari: senin

Hasil pencarian jadwal hari senin:
==========================================================================
Nama Mata Kuliah         Ruang          Hari Kuliah         Jam Kuliah
==========================================================================
daspro                   rt1\           senin               8 - 12
==========================================================================
pilih menu:     1: input data   2: tabel jadwal        3: cari data sesuai hari       4: cari data sesuai matakuliah
===4
Masukkan nama mata kuliah yang dicari: daspro

Hasil pencarian mata kuliah: daspro
==========================================================================
Nama Mata Kuliah         Ruang          Hari Kuliah         Jam Kuliah
==========================================================================
daspro                   rt1\           senin               8 - 12
==========================================================================
pilih menu:     1: input data   2: tabel jadwal        3: cari data sesuai hari       4: cari data sesuai matakuliah
```